

CLINTON, MO, USA

WMO: 720636

Lat: 38.350N

Lon: 93.683W

Elev: 824

StdP: 14.26

Time Zone: -6.00 (NAC)

Period: 09-19

WBAN: 00222

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.0	10.3	-3.8	4.6	7.2	1.1	6.0	13.2	24.2	34.5	21.5	34.9	4.2	10	0.459

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	97.0	76.5	92.9	77.2	90.6	77.0	80.9	90.0	79.5	88.8	78.2	87.5	7.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.9	154.9	86.9	76.9	144.7	85.5	75.1	136.0	83.4	45.4	88.6	43.7	88.3	42.1	88.2	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.2	18.0	16.1	-1.6	100.2	8.9	5.6	-8.0	104.2	-13.2	107.5	-18.1	110.6	-24.6	114.7		
			-2.5	83.1	8.5	1.9	-8.6	84.5	-13.6	85.6	-18.4	86.7	-24.6	88.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.6	32.1	34.5	47.2	56.9	66.1	76.2	78.9	76.4	69.9	57.3	45.9	36.7
	DBStd	18.76	11.54	11.53	11.53	8.95	8.67	5.80	6.03	5.83	7.54	9.68	10.19	9.82
	HDD50	1885	562	445	188	36	4	0	0	0	0	38	188	424
	HDD65	4568	1020	853	561	267	98	2	0	2	35	278	574	878
	CDD50	4305	7	12	102	243	502	785	896	820	596	265	66	11
	CDD65	1512	0	0	10	24	131	337	431	357	181	39	2	0
	CDH74	15246	0	0	67	226	1097	3482	4869	3546	1619	329	11	0
	CDH80	6396	0	0	6	37	298	1457	2330	1572	619	76	1	0
Wind	WSAvg	6.4	7.2	7.6	8.3	8.3	6.5	6.2	4.8	4.2	4.6	6.0	7.0	6.8
Precipitation	PrecAvg	43.7	1.8	2.1	2.9	4.9	5.5	5.5	4.3	3.5	4.4	3.4	3.2	2.3
	PrecMax	58.5	6.0	5.6	6.2	15.8	13.2	10.3	13.9	7.4	13.0	9.2	10.8	7.1
	PrecMin	28.9	0.3	0.2	0.9	1.6	1.4	1.7	0.6	0.7	1.3	0.5	0.7	0.5
	PrecStd	7.8	1.2	1.4	1.5	2.6	2.6	2.4	3.1	1.7	2.8	2.2	2.5	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.4	70.4	81.0	84.0	90.0	96.8	102.2	100.0	94.8	85.9	75.2	68.4
		MCWB	56.2	55.5	65.8	68.1	74.6	78.2	75.9	74.7	75.1	73.8	61.6	60.5
	2%	DB	60.8	63.7	74.9	80.7	86.0	92.5	98.7	95.0	90.4	81.9	70.4	62.8
		MCWB	51.2	54.8	62.1	64.8	72.3	77.9	76.0	76.6	75.9	67.3	59.7	55.5
	5%	DB	54.4	57.4	71.6	75.4	83.5	90.5	93.4	91.1	87.5	77.3	65.9	56.6
		MCWB	46.5	48.9	59.0	62.4	71.9	77.3	77.7	76.3	74.7	66.6	57.3	49.9
	10%	DB	48.1	52.2	65.6	72.5	81.1	88.4	90.8	88.4	82.5	73.1	62.7	52.4
		MCWB	42.6	44.7	55.9	61.1	70.0	76.4	77.9	75.5	72.2	63.3	54.3	47.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.0	61.3	67.7	70.8	76.9	81.8	83.2	83.2	78.5	74.2	65.2	62.7
		MCDB	63.0	67.5	78.2	81.0	86.6	90.5	91.7	93.3	88.3	83.2	71.5	66.7
	2%	WB	52.0	56.8	64.4	67.4	74.7	80.0	81.1	80.3	77.1	71.5	61.9	56.3
		MCDB	58.5	62.0	73.1	75.9	82.9	89.5	90.0	89.5	87.4	78.1	66.5	60.4
	5%	WB	47.2	49.7	60.9	65.1	72.7	78.4	79.7	78.4	75.6	68.6	58.8	51.7
		MCDB	52.9	59.0	69.0	73.2	80.9	88.1	88.9	87.1	84.7	74.0	64.7	55.3
	10%	WB	42.6	44.8	56.9	62.9	71.1	77.2	78.3	77.1	73.9	65.6	55.1	46.7
		MCDB	48.7	51.0	62.2	69.6	79.0	86.6	87.6	85.7	81.1	72.1	61.3	50.8
Mean Daily Temperature Range	5% DB	MDBR	18.5	18.0	20.1	21.1	19.3	18.9	19.3	19.9	20.8	22.1	20.3	16.9
		MCDBR	26.0	26.6	26.2	25.4	21.7	20.1	22.2	23.6	22.5	24.6	25.6	24.4
		MCWBR	17.7	19.0	14.9	14.3	11.3	9.7	7.7	9.5	10.1	13.0	17.3	17.3
	5% WB	MCDBR	23.9	25.4	22.2	20.7	19.4	18.9	18.1	19.2	20.3	19.2	22.2	21.6
		MCWBR	17.8	19.9	14.0	13.7	11.0	10.0	8.9	9.7	10.2	11.7	17.3	18.2
Clear-Sky Solar Irradiance	taub	0.300	0.313	0.339	0.387	0.428	0.431	0.446	0.449	0.389	0.346	0.309	0.294	
	taud	2.459	2.429	2.410	2.290	2.238	2.264	2.246	2.236	2.364	2.469	2.515	2.503	
	Ebn at Noon	280	291	292	283	272	270	265	261	272	275	275	275	
	Edh at Noon	26	30	34	41	44	43	43	43	36	29	24	23	
All-Sky Solar Radiation	RadAvg	710	928	1221	1565	1800	2048	2038	1802	1532	1100	763	607	
	RadStd	79	116	104	99	159	165	127	107	102	117	81	59	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (26 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page